

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)	(i)		Expanding	1			1		
		(ii)		A single point	1			1		
	(b)	(i)		Any 2 ×(1) from: - same number in both - same pattern or accept same distance apart - red shifted <u>in B</u> / shifted to the right <u>in B</u> / longer wavelengths <u>in B</u>. Accept closer to the left <u>in A</u> or shorter wavelengths <u>in A</u>			2	2		
		(ii)		X is further away (1) X is moving faster (1) Accept converse and diagram B for X			2	2		
				Question 8 total	2	0	4	6	0	0